# Launch Control 3

**PERFORMANCE MAP · WORKING SOURCE**
16 endless encoders · 8 buttons · no faders
Compact LC3 — not Launch Control XL 3

**FEEDBACK REQUIRED.** AP OSC input: **9090** · feedback: **127.0.0.1:9096** · **Toggle All** checked. Missing feedback blocks EQ/filter pickup and Mute/Cue toggles.

**FUNCTION · BUTTON LAYER**
Toggle Hot Cues ↔ Mixer
Startup: Hot Cues / Deck 1

**TRACK ◀ / ▶ · DECK 1–4**
Select hot-cue target · no wrap
Screen shows deck and button layer

**BROWSER · TOP RIGHT**
Clockwise: Down · anticlockwise: Up
Same encoder in both modes

**EQ / FILTER ENCODERS**
Mode + Function (DAW)
Always fixed mixer CH1–4

**Read each row left → right, positions 1–8.** The two diagrams are alternative modes of the same surface. "—" means unassigned.

## MODE 1 · DAW MIXER → EQ

Console mode: eq · startup requests this mode

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| TOP | LOW CH1 | MID CH1 | HIGH CH1 | LOW CH2 | MID CH2 | HIGH CH2 | — unassigned | BROWSE Up / Down |
| BOTTOM | LOW CH3 | MID CH3 | HIGH CH3 | LOW CH4 | MID CH4 | HIGH CH4 | — unassigned | — unassigned |

## MODE 2 · DAW CONTROL → FILTERS

Console mode: filter · filter input routes unconfirmed

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| TOP | FILTER 1 CH1 | FILTER 2 CH1 | FILTER 1 CH2 | FILTER 2 CH2 | — unassigned | — unassigned | — unassigned | BROWSE Up / Down |
| BOTTOM | FILTER 1 CH3 | FILTER 2 CH3 | FILTER 1 CH4 | FILTER 2 CH4 | — unassigned | — unassigned | — unassigned | — unassigned |

**DEFAULT · HOT CUES** — bottom buttons 1–8 launch the selected deck's cue slots. Function toggles to/from the Mixer layer below.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| HOT CUE 1 | HOT CUE 2 | HOT CUE 3 | HOT CUE 4 | HOT CUE 5 | HOT CUE 6 | HOT CUE 7 | HOT CUE 8 |

**FUNCTION → MIXER LAYER** — existing mute/headphone-cue controls, unchanged by Track selection.

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| MUTE CH1 | MUTE CH2 | MUTE CH3 | MUTE CH4 | CUE CH1 | CUE CH2 | CUE CH3 | CUE CH4 |

**Hot-cue colours:** fixed Candy Pop RGB palette, NOT populated-cue indicators. Empty slots can set cues. Function: purple = Hot Cues; cyan = Mixer.

**Mixer LEDs:** red = mute; cyan = headphone cue. Bright/dim follows AP on/off feedback; grey = unknown. Knobs are unchanged by the button layer.

# Setup & feedback

One bridge · shared AP connection
Use alongside FLX4 or on its own

## 1 · Configure AUTO/PILOT

| Setting | Value |
|---|---|
| OSC input / listen | **Enabled · 9090** |
| OSC feedback destination | **127.0.0.1 · 9096** |
| Feedback update rate | **30 Hz** |
| Feedback subscriptions | **Toggle All → all checked** |
| Browser Mode | **CONTROL MODE** |
| Direct AP MIDI inputs | **Disable LC3** and any other bridged controller |

For LC3-only selective feedback, enable CH1–4 `EQ/Low`, `EQ/Mid`, `EQ/High`, `Filter1`, `Filter2`, `Mute` and `Cue`. The FLX4 needs additional subscriptions.

## 2 · Use the DAW USB ports

Close other apps holding LC3's DAW ports. On Windows the documented DAW connection is **USB MIDI port 2**, not Main.

| | |
|---|---|
| **Input** | `MIDIIN2 (LC3 1 MIDI)` |
| **Output** | `MIDIOUT2 (LC3 1 MIDI)` |

These are the development machine's names. Run `ap-midi-osc-bridge.exe --list` to list yours without sending MIDI. If different, edit `inputs` / `outputs` in `profiles/launch-control-3.json`. Input/output indices are independent; selectors must match uniquely.

**Do not select:** Main `LC3 1 MIDI` or an LCXL3 port.

## Modes & encoder pickup

Press/hold **Mode** and press **Function (DAW)** to switch DAW Mixer ↔ DAW Control, then exit mode selection. Other device modes are not mapped.

A confirmed change of mode re-arms pickup. AP's reported EQ/filter positions are sent back to the endless encoders over the DAW output, so they can start from the current value rather than jump blindly. **No position feedback → no EQ/filter output.**

**Function alone** toggles the bottom row Hot Cues ↔ Mixer without changing the EQ/filter mode or encoder pickup. Track ◀/▶ selects Deck 1–4 for hot cues, independently of FLX4. The screen shows deck/layer; arrows stop at the ends.

Knobs and Mixer-layer buttons remain fixed to **CH1–4**, regardless of either controller's deck selection. These are AP's main mixer channels, not arrangement-internal trims.

## Browser & button behaviour

**Top encoder 8:** clockwise moves Down; anticlockwise moves Up. The bridge recentres its absolute value after turns for continuous browsing. After a mode change, the first non-echo report establishes a baseline and may consume the first turn.

Matching reports can be suppressed within a 500 ms echo window, including an identical physical report. This is not a MIDI acknowledgment guarantee.

**Hot Cues:** one tap per press, no held preview. Colours match the supplied Candy Pop palette, but do not show cue existence or follow later AP colour changes. Empty slots can create cues; AP's existing CueMomentary/Slip settings are not changed. Track routes can act as arrangement markers, not clip/sampler cues.

**Mixer-layer Mute / headphone Cue:** a press sends the opposite of AP's confirmed 0/1 state. LEDs follow feedback, not a guessed local toggle. Grey/unknown means the bridge will not guess a switch state.

LC3 has no mapped Browser Enter, Back or Load. Use AP or the FLX4 for those actions.

> **FILTER ROUTES ARE NOT CONFIRMED.**
> `/AutoPilot/Mixer/CHn/Filter1` and `Filter2` are inferred from AP's 0.1.845 change note and mixer path convention. Missing feedback blocks pickup; feedback alone does not prove the input route is writable.

## 3 · Start the shared bridge

Keep `profiles/` beside the current executable. The bridge automatically matches connected controllers to their profiles; absent devices are skipped. Connect only LC3 for LC3-only use. No device-list config or Components remapping is needed.

On Windows, double-click **ap-midi-osc-bridge.exe**; on macOS/Linux, run `./ap-midi-osc-bridge`. No command launcher is needed. **Run only one bridge instance.**

Startup requests DAW Mixer (EQ), with the **Hot Cues** button layer and **Deck 1**. The controller must report a mapped DAW mode before controls become active. No firmware or stored Custom Modes are overwritten.

## If controls do nothing

Check the console mode, DAW input **and** output selection, AP feedback destination/subscriptions, and Browser CONTROL MODE. The bridge does not configure AP for you.

Keep the console open. **Ctrl+C** requests standalone mode and releases held actions best-effort. Device attach/detach is handled while running; check console errors. Restart after restarting AP or uncertain state. Force-kills and UDP loss cannot guarantee cleanup.

**Local references:** README.md → Get started / Supported mappings / Limits; profiles/launch-control-3.json → modes / layers / deck_feedback / bindings / feedback; PROFILES.md → Declarative JSON / Protocol references.
**Scope:** current default profile, not a claim about an older FLX4 Pilot package. Hardware and AP end-to-end operation remain unverified. Edited profiles may differ from this sheet.